Dynamics – Project 4: Spring Constant

Objective

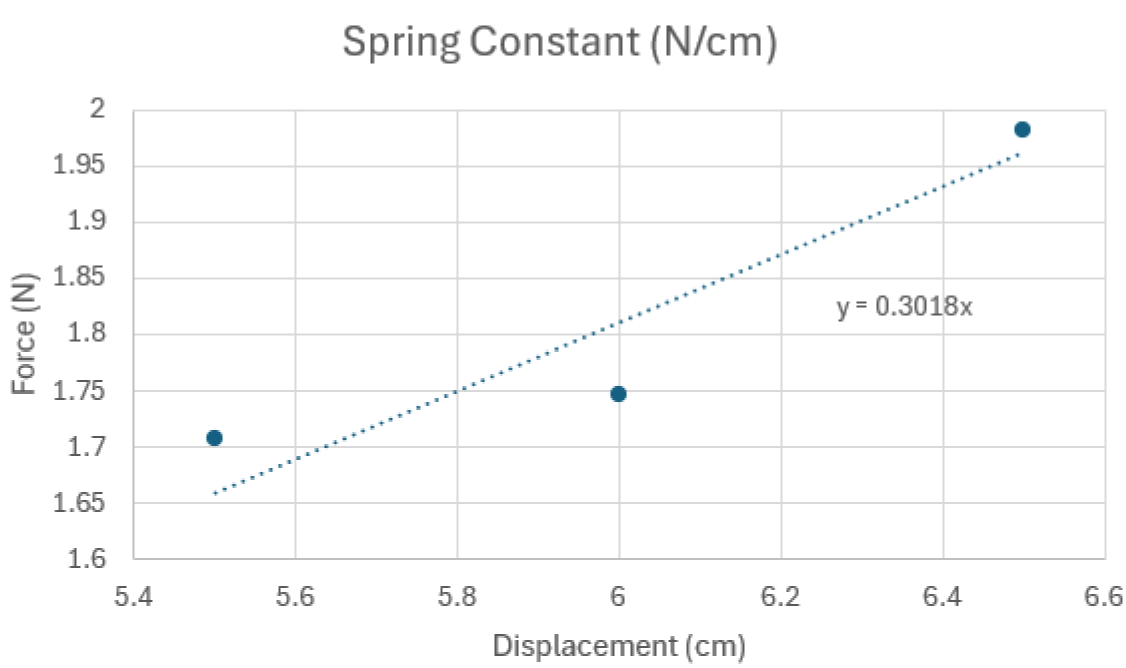
For this project, we were in a group of 3 and the objective was to measure the constant of a spring. According to the documentation we can use the formula: $F = k * \Delta x$, the objective is to identify the constant k by doing several measurements.

Methodology

To measure the constant k, we decided to use a rubber band because we didn't have a spring. We also used a tape measure to measure the elongations of the rubber band. And for the objects, we used our phones. Because it was very easy to determine the weight of them just by checking on the internet. To obtain the measurements, we put the rubber band around the phone, we put all this device on a table, then we lift the phone from the table by pulling the rubber band. And we use the tape measure to obtain the elongation. With all of this, and with the formula: $F = k * \Delta x$, it's very easy to have the constant k. We obtained three measurements of k because we use our three phones.

Analysis

Mass (g)	Gravitational Constant	Force (N)	Unstretched (cm)	Stretched (cm)	Displacement (cm)	Spring Constant (N/m)	Error%
174	9.81	1.706	2	7.5	5.5	31.035	2.738
202	9.81	1.981	2	8.5	6.5	30.486	0.921
178	9.81	1.746	2	8	6	29.103	3.659
Average	Average	Average	Average	Average	Average	Average	
184.67	9.81	1.811	2	8	6	30.208	



Trendline matches expected value. It is based on N/cm, so it would be 30.18 N/m.

Hand computational work

Project 4 Brainstorm

rubber band ⇒

$$\bar{F} = k \cdot \Delta X$$

$$k = \frac{\bar{F}}{\Delta X} = \frac{mg}{(x_2 - x_1)}$$

(1) iphone 13 ⇒ 174 g = 0.174 Kg ⇒ $(0.174\ \text{kg})(9.81\ \frac{m}{s^2})$

$= 1.707\ N$

2 cm unstretch
2.5 cm stretch

$$k = \frac{1.707\ N}{(2.5 \cdot 10^{-2} - 2 \cdot 10^{-2})} = \boxed{31.04\ \frac{N}{m}}$$

0.025 m − 0.02 m = 0.0

Hand computational work continued

(2) Samsung galaxy A54 = 202 g = 0.202 kg

8.5 cm stretched
2 cm unstretched

$$K = \frac{F}{\Delta x} = \frac{mg}{x_2 - x_1} = \frac{(0.202\ kg)(9.81\ \frac{m}{s^2})}{8.5\ cm - 2\ cm} = \boxed{30.49\ \frac{N}{m}}$$

(3) 8 cm   178 g
2 cm

$$\frac{(0.178\ kg)(9.81\ \frac{m}{s^2})}{8\ cm - 2\ cm} = \boxed{29.1\ \frac{N}{m}}$$

Video:

https://youtube.com/shorts/5-HBDCsp6Uk?si=dE5ewuKatHYDrW3z

Conclusion

Finally, we obtained three close values (31.04 N/m, 30.49 N/m and 29.1 N/m). Our graph is almost linear too, so we can say that the results seem pretty good. To increase the accuracy of the result we can think about a better way to attach the phone to the rubber band. And the method that we used to measure the elongation of the rubber band. Because we have some uncertainty with the method that we just used.

Picture:

looks out for the preservation of the universe.
number one: Learn to live with each other. Number two: try to bring
it takes a hundred years, it will be a pity, but I will not cease to
he whole art of making experiments in chemistry is founded
exact equality or equation between the
Lavoisier
Good
exercise
us try to
your
Stephanie
Greenfield
Life is an